Chapter 3 Vector Spaces 向量空间

Section 3.1 Definition and examples

Definition (Vector space) Let V be a set of objects (called *elements*) on which the operations of *addition* and *scalar multiplication* are defined that satisfy the two closure properties and eight axioms below.

C1 For each $\mathbf{x}, \mathbf{y} \in V$, $\mathbf{x} + \mathbf{y} \in V$

C2 For each $\alpha \in \mathbf{R}$ and each $\mathbf{y} \in V$, $\alpha\mathbf{y} \in V$.

A1 $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$, for all $\mathbf{x}, \mathbf{y} \in V$;

A2 $\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z}$, for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$;

A3 There exists a zero element $\mathbf{0} \in V$ such that $\mathbf{x} + \mathbf{0} = \mathbf{x}$ for all $\mathbf{x} \in V$;

A4 For each element $\mathbf{x} \in V$, there exists an additive inverse $\mathbf{y} \in V$ such that $\mathbf{x} + \mathbf{y} = \mathbf{0}$;

A5 $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$ for each scalar α and any $\mathbf{x}$ and $\mathbf{y}$ in V.

A6 $(\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$ for any scalars α and β and any $\mathbf{x}$ in V.

A7 $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$ for any scalars α and β and any $\mathbf{x} \in V$.

A8 $1\mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in V$.

关于封闭性质 8 个公理.

① C_1 对 $x, y \in V$. $x+y \in V$

相加仍在 V 中.

C_2. $\alpha \in \mathbb{R}$. $y \in V$. $\alpha y \in V$

② A_1 $x+y = y+x$. Commutativity 交换

A_2 $x+(y+z) = (x+y)+z$ ($x, y, z \in V$) associativity 结合

A_3 $0 \in V$. $\Rightarrow$ every $x \in V$. $x+0 = 0$

A_4 $y \in V$. $\rightarrow x+y = 0$

A_5 $\alpha(x+y) = \alpha x + \alpha y$ distributive 分配律.

A_6 $(\alpha+\beta)x = \alpha x + \beta x$ for each scalars α and β ($\mathbb{R}$. ②)

A_7 $(\alpha\beta)x = \alpha(\beta x)$

A_8 $1 \cdot x = x$. $x \in V$

欧几里得

Example The Vector Space $\mathbf{R}^n$ (Euclidean Vector Space)

Let $\mathbf{R}^n = \{(a_1, a_2, a_3, \cdots, a_n)^T | a_i \in \mathbf{R}\}$.

Define the standard addition and scalar multiplication on $\mathbf{R}^n$ by

$$(a_1, a_2, \cdots, a_n) + (b_1, b_2, \cdots, b_n) = (a_1 + b_1, a_2 + b_2, \cdots, a_n + b_n)$$

and

$$\alpha(a_1, a_2, \cdots, a_n) = (\alpha a_1, \alpha a_2, \cdots, \alpha a_n)$$

for any $(a_1, a_2, \cdots, a_n), (b_1, b_2, \cdots, b_n) \in \mathbf{R}^n$ and $\alpha \in \mathbf{R}$.

$\mathbf{R}^n$ is a vector space. In this case, the zero vector is $\vec{0} = (0, \cdots, 0)$.

Notation ($\mathbf{e}_i$) Denote $\mathbf{e}_i$ to be the vector with entry 1 in the ith position and entries 0 in other positions.

Example The Vector Space P_n

Let P_n be the set of all polynomials of degree less than n. That is, $P_n = \{a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0 | a_i \in \mathbf{R}\}$.

Define the standard addition and scalar multiplication of polynomials by

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots (a_{n-1} + b_{n-1})x^{n-1}$$

and

$$\alpha p(x) = \alpha a_0 + \alpha a_1 x + \cdots + \alpha a_{n-1}x^{n-1}$$

for any $p(x) = a_0 + \cdots + a_{n-1}x^{n-1}$, $q(x) = b_0 + \cdots + b_{n-1}x^{n-1} \in P_n$ and $\alpha \in \mathbf{R}$.

Then P_n is a vector space.

$$p(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}$$

$$q(x) = b_0 + b_1 x + \cdots + b_{n-1} x^{n-1}$$

加.来源是后面定义的.

define "+" as

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_{n-1} + b_{n-1})x^{n-1}$$

define "*"

$$\alpha p(x) = (\alpha a_0) + (\alpha a_1)x + \cdots + (\alpha a_{n-1})x^{n-1}$$

Example

Let $\mathbf{R}^{m\times n}$ be the set of all $m \times n$ matrices with real entries.

The addition and scalar multiplication in $\mathbf{R}^{m\times n}$ are defined as the usual matrix addition and scalar multiplication defined in Section 1.3.

We have check in Section 1.3 that $\mathbf{R}^{m\times n}$ satisfies all axioms of a vector space, and hence is a vector space. In particular, the additive identity is the zero matrix.

Example

Let V be the set of infinite sequences. That is $V = \{(a_1, a_2, a_3, \cdots) | a_i \in \mathbf{R}\}$.
For example, $(1, -1, 1, -1, \cdots)$, $(1, 2, 3, 4, \cdots)$,$(1, 4, 7, 11, \cdots)$ are in V.

Define $\mathbf{a} + \mathbf{b}$ and $\alpha\mathbf{a}$ respectively by

$$(\mathbf{a} + \mathbf{b})_i = a_i + b_i \quad \text{and} \quad (\alpha\mathbf{a})_i = \alpha a_i$$

For example, $2(1, -1, 1, -1, \cdots) = (2, -2, 2, -2, \cdots)$

$$(1, 2, 3, 4, \cdots) + (1, 1, 1, 1, \cdots) = (2, 3, 4, 5, \cdots).$$

In this case, the zero vector is $(0, 0, 0, 0, \cdots)$.
It can be verified that all the vector space axioms hold. V is a vector space.

Example

Let $C[a, b]$ be the set of all continuous functions with domain $[a, b]$ and range $\mathbf{R}$.

The sum $f + g$ of two functions in $C[a, b]$ is defined by

$$(f + g)(x) = f(x) + g(x)$$

for all x in $[a, b]$.

For any scalar α, the scalar multiplication αf in $C[a, b]$ is defined by

$$(\alpha f)(x) = \alpha f(x)$$

for all x in $[a, b]$.

Then $C[a, b]$ is a vector space.

Example Let $V = \{(a, 1) : a \in \mathbf{R}\} \subset \mathbf{R}^2$. Define a new addition $+_{new}$ and scalar multiplication $\cdot_{new}$ of V by

$$(a, 1) +_{new} (b, 1) := (a + b, 1)$$

and

$$\alpha \cdot_{new} (a, 1) := (\alpha a, 1)$$

for any $(a, 1), (b, 1) \in V$ and $\alpha \in \mathbf{R}$.

A vetor space is a set equipped with two calculations. Satisfying some properties.

一个集合上
规定2种(加法)的计算 ⇒ vector space

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$$(a, 1) +_{new} (b, 1) := (a + b, 1)$$

and

$$\alpha \cdot_{new} (a, 1) := (\alpha a, 1)$$

for any $(a, 1), (b, 1) \in V$ and $\alpha \in \mathbf{R}$.

In this case, the zero vector is $(0, 1)$, but not $(0, 0)$.

All the vector space axioms hold.

Thus, V over $\mathbf{R}$, with $+_{new}$ and $\cdot_{new}$, is a vector space.

new definition

Example Let $V = \{(x, y)^T | x, y \in \mathbf{R}\}$. Define addition componentwise, that is

$$(x_1, x_2)^T + (y_1, y_2)^T = (x_1 + y_1, x_2 + y_2)^T$$

and define scalar multiplication by $\alpha \in \mathbf{R}$ to be

$$\alpha(x_1, x_2)^T = (\alpha x_1, 0)^T.$$

Determine whether or not V under these operations is a vector space.

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and define scalar multiplication by $\alpha \in \mathbf{R}$ to be

$$\alpha(x_1, x_2)^T = (\alpha x_1, 0)^T.$$

Determine whether or not V under these operations is a vector space.

Solution Let $\mathbf{x} = (2, 3)^T$. By the definition of scalar multiplication, $1\mathbf{x} = 1(2, 3)^T = (1 \times 2, 0)^T = (2, 0)^T \neq \mathbf{x}$. So axiom A8 (the existence of a multiplicative identity) does not hold. V is not a vector space under these prescribed operations.

Example Let V denote the set of all polynomials of degree n, $n \geq 1$,

$$V = \{a_0 + a_1 x + \cdots + a_n x^n : a_0, \cdots, a_n \in \mathbf{R},\ a_n \neq 0\}.$$

Define the standard addition and scalar multiplication of polynomials by

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots (a_n + b_n)x^n$$

and

$$\alpha p(x) = \alpha a_0 + \alpha a_1 x + \cdots + \alpha a_n x^n$$

for any $p(x) = a_0 + \cdots + a_n x^n$, $q(x) = b_0 + \cdots + b_n x^n \in V$ and $\alpha \in \mathbf{R}$. Determine whether or not V under these operations is a vector space.

Solution

Since $(1 + x^n) + (-x^n) = 1 \notin V$, the addition is not closed. That is, C1 is not satisfied. So V, with the standard addition and scalar multiplication of polynomials, is NOT a vector space.

Theorem If V is a vector space and $\mathbf{x}$ is any element of V, then

(i) $0\mathbf{x} = \mathbf{0}$

(ii) If $\mathbf{x} + \mathbf{y} = \mathbf{0}$ and $\mathbf{x} + \mathbf{z} = \mathbf{0}$, then $\mathbf{y} = \mathbf{z}$. That is, the additive inverse of $\mathbf{x}$ is unique.

(iii) $\mathbf{x} + (-1)\mathbf{x} = \mathbf{0}$. That is, the additive inverse of $\mathbf{x}$ is $(-1)\mathbf{x}$.

$\neq x[-x]$ inverse of x

Proof A proof can be found in the textbook.

Example In $\mathbf{R}^3$,

(i) $0(1,2,3)^T = (0,0,0)^T$

(ii) If $(1,2,3)^T + (y_1,y_2,y_3)^T = (0,0,0)^T$ and $(1,2,3)^T + (z_1,z_2,z_3)^T = (0,0,0)^T$, then $(y_1,y_2,y_3)^T = (z_1,z_2,z_3)^T$.

(iii) $(1,2,3)^T + (-1)(1,2,3)^T = (1,2,3)^T + (-1,-2,-3)^T = (1-1,2-2,3-3)^T = (0,0,0)^T$.

Remark Since the additive inverse of $\mathbf{x}$ is unique, we write $-\mathbf{x}$ for the additive inverse of $\mathbf{x}$.

Theorem 3.1.1 *If V is a vector space and $\mathbf{x}$ is any element of V, then*

(i) $0\mathbf{x} = \mathbf{0}$.

(ii) $\mathbf{x} + \mathbf{y} = \mathbf{0}$ *implies that* $\mathbf{y} = -\mathbf{x}$ *(i.e., the additive inverse of* $\mathbf{x}$ *is unique).*

(iii) $(-1)\mathbf{x} = -\mathbf{x}$.

Proof It follows from axioms A6 and A8 that

$$\mathbf{x} = 1\mathbf{x} = (1 + 0)\mathbf{x} = 1\mathbf{x} + 0\mathbf{x} = \mathbf{x} + 0\mathbf{x}$$

Thus,

$$-\mathbf{x} + \mathbf{x} = -\mathbf{x} + (\mathbf{x} + 0\mathbf{x}) = (-\mathbf{x} + \mathbf{x}) + 0\mathbf{x} \qquad \textbf{(A2)}$$

$$\mathbf{0} = \mathbf{0} + 0\mathbf{x} = 0\mathbf{x} \qquad \textbf{(A1, A3,} \text{ and } \textbf{A4)}$$

To prove **(ii)**, suppose that $\mathbf{x} + \mathbf{y} = \mathbf{0}$. Then

$$-\mathbf{x} = -\mathbf{x} + \mathbf{0} = -\mathbf{x} + (\mathbf{x} + \mathbf{y})$$

Therefore,

$$-\mathbf{x} = (-\mathbf{x} + \mathbf{x}) + \mathbf{y} = \mathbf{0} + \mathbf{y} = \mathbf{y} \qquad \textbf{(A1, A2, A3,} \text{ and } \textbf{A4)}$$

Finally, to prove **(iii)**, note that

$$\mathbf{0} = 0\mathbf{x} = (1 + (-1))\mathbf{x} = 1\mathbf{x} + (-1)\mathbf{x} \qquad \textbf{[(i) and A6]}$$

Thus,

$$\mathbf{x} + (-1)\mathbf{x} = \mathbf{0} \qquad \textbf{(A8)}$$

and it follows from part **(ii)** that

$$(-1)\mathbf{x} = -\mathbf{x}$$

■

Chapter 3 Vector Spaces

Section 3.2 Subspaces

Definition (Subspace) A subset S of a vector space V over $\mathbf{R}$ is said to be a *subspace* of V if:

(i) S is non-empty, i.e. S contains at least an element.

(ii) $\mathbf{u} + \mathbf{v} \in S$ for any vectors $\mathbf{u}, \mathbf{v} \in S$.

(iii) $\alpha\mathbf{v} \in S$ for any vector $\mathbf{v} \in S$ and scalar $\alpha \in \mathbf{R}$;

→ a vector space as well (itself)

Definition (Subspace) A subset S of a vector space V over $\mathbf{R}$ is said to be a *subspace* of V if:

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(ii) $\mathbf{u} + \mathbf{v} \in S$ for any vectors $\mathbf{u}, \mathbf{v} \in S$.

(iii) $\alpha\mathbf{v} \in S$ for any vector $\mathbf{v} \in S$ and scalar $\alpha \in \mathbf{R}$;

Remark Any subspace of a vector space is a vector space itself.

Reason

- (ii) and (iii) are C1, C2.
- A1, A2, A5, A6, A7, A8 hold for all vectors in V, they also hold for the vectors in S.
- Substitute $\alpha = 0$ in (iii), we have A3.
- Substitute $\alpha = -1$ in (iii), we have A4.

Example Show that $S = \left\{ (a_1, a_2, a_3)^T \in \mathbf{R}^3 : \ a_1 + a_2 + a_3 = 0 \right\}$ is a subspace of $\mathbf{R}^3$.

① $S \neq \phi$ (不是空集)

②③ 封闭性检验.

Example Show that $S = \left\{ (a_1, a_2, a_3)^T \in \mathbf{R}^3 : \ a_1 + a_2 + a_3 = 0 \right\}$ is a subspace of $\mathbf{R}^3$.

$\Rightarrow \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ in column (为了考虑空间)

Solution S is a subset of a vector space $\mathbf{R}^3$.

at least one exists.

(i) Since the zero vector $(0, 0, 0)^T \in S$, S is nonempty.

(ii) Let $(a_1, a_2, a_3)^T \in S$, $\alpha \in \mathbf{R}$. Then $a_1 + a_2 + a_3 = 0$. Since $\alpha a_1 + \alpha a_2 + \alpha a_3 = 0$,

prove the defined "*"

$$\Rightarrow \alpha (a_1, a_2, a_3)^T = (\alpha a_1, \alpha a_2, \alpha a_3,)^T \in S.$$

(iii) Let $\mathbf{a}, \mathbf{b} \in S$. Then $a_1 + a_2 + a_3 = 0$ and $b_1 + b_2 + b_3 = 0$. Since $(a_1 + b_1) + (a_2 + b_2) + (a_3 + b_3) = 0$,

a.b ∈ S

$$\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2, a_3 + b_3)^T \in S.$$

Since S is nonempty and satisfies the two closure conditions, S is a subspace of $\mathbf{R}^3$.

Let $\vec{a} = (a_1, a_2, a_3)^T$ $\quad \vec{a} + \vec{b} = (a_1+b_1, a_2+b_2, a_3+b_3)$

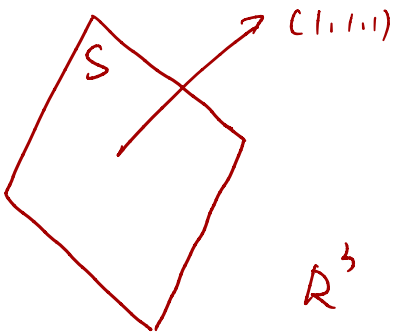
$\vec{b} (b_1, b_2, b_3)^T$

want to prove : $(a_1+b_1)+(a_2+b_2)+(a_3+b_3) = 0$

$\Rightarrow (a_1+a_2+a_3)+(b_1+b_2+b_3) = 0+0 = 0$

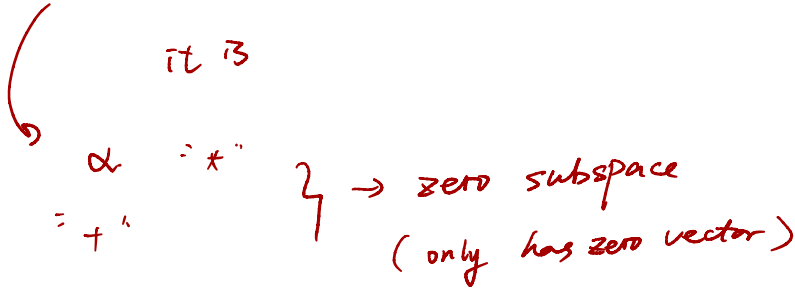
$\Rightarrow$ S is a subspace of R^3 → 三维空间

↳ 平面 (包含 $(0,0,0)$) → 过原点的平面都是 subspace



If $a_1+a_2+a_3 = 1$ $\Rightarrow$ S can't be a subspace

Example Let $S = \{(0,0,0)^T\}$. Is S a subspace of $\mathbf{R}^3$?



Example Let $S = \{(0,0,0)^T\}$. Is S a subspace of $\mathbf{R}^3$?

Solution

(i) The set S is nonempty, since $x = (0,0,0)^T \in S$.

(ii) Let $\mathbf{x} \in S$ and $\alpha \in \mathbf{R}$. Then $\mathbf{x} = (0,0,0)^T$. Since $\alpha\mathbf{x} = \alpha(0,0,0)^T = (0,0,0)^T \in S$, $\alpha\mathbf{x} \in S$.

(iii) Let $\mathbf{x}, \mathbf{y} \in S$. Then $x = (0,0,0)^T$ and $\mathbf{y} = (0,0,0)^T$. So, $\mathbf{x} + \mathbf{y} = (0,0,0)^T + (0,0,0)^T = (0,0,0)^T \in S$.

Since S is nonempty and satisfies the two closure conditions, S is a subspace of $\mathbf{R}^3$.

Definition (Zero subspace) For any vector space V, $\{\mathbf{0}\}$ is a subspace of V. This subspace is called the zero subspace of V.

Definition (Trivial subspaces, proper subspaces) For any vector space V, $\{\mathbf{0}\}$ and V are subspaces of V. Both $\{\mathbf{0}\}$ and V are called the *trivial subspaces*. All other subspaces are referred to as *proper subspaces*.

平凡子空间

Example Let $S = \{A \in \mathbf{R}^{n\times n} | A \text{ is a symmetric matrix}\}$. Show that S is a subspace.

对称矩阵

$A^T = A$

n^2 dimensions

$\dim V = n^2$

子空间

$V = \mathbb{R}^{n\times n}$

$S = \{A \in \mathbb{R}^{n\times n} : A^T = A\}$

(i) $0_{n\times n} \in S$ (at least one vector) $S \neq \emptyset$

(ii) $A \in S$, $B \in S$ $\Rightarrow$ Show $A+B \in S$

$(A+B)^T = A^T + B^T = A + B \Rightarrow (A+B)^T = (A+B) \Rightarrow A+B \in S$

(iii) $A \in S$, $\alpha \in \mathbb{R}$ $\Rightarrow$ show $\alpha A \in S$

$(\alpha A)^T = \alpha A^T = \alpha A \Rightarrow \alpha A \in S$.

from (i)(ii)(iii)

we can have S is a subspace

Example Let $S = \{A \in \mathbf{R}^{n\times n} | A \text{ is a symmetric matrix}\}$. Show that S is a subspace.

Solution

(i) The set S is nonempty, since $I \in S$.

(ii) Let $A \in S$ and $\alpha \in \mathbf{R}$. Then $A^T = A$. Since

$$(\alpha A)^T = \alpha A^T = \alpha A,$$

$\alpha A \in S$.

(iii) Let $A, B \in S$. Then $A = A^T$ and $B = B^T$. Since

$$(A + B)^T = A^T + B^T = A + B,$$

$A + B \in S$.

Since S is nonempty and satisfies the two closure conditions, S is a subspace of $\mathbf{R}^{n\times n}$.

零空间

Definition (Null space) Given $A_{m\times n}$, let $N(A)$ denote the set of all solutions of system of linear equations $A\mathbf{x} = \mathbf{0}$, that is,

$$N(A) = \{x \in \mathbf{R}^n : A\mathbf{x} = \mathbf{0}\},$$

which is called the *null space* of A.

① $\vec{x} \in N(A)$. $\vec{y} \in N(A)$. Show: $\vec{x}+\vec{y} \in N(A)$

$A(\vec{x}+\vec{y}) = A\vec{x}+A\vec{y} = 0+0 = 0$

$\Rightarrow A(\vec{x}+\vec{y}) = 0 . (\vec{x}+\vec{y}) \in N(A)$

① $\vec{0}$ is a solution

② Let $\vec{x} \in N(A)$. $\alpha \in \mathbb{R}$

Show: $A(\alpha\vec{x}) = 0$

$A(\alpha\vec{x}) = \alpha(A\vec{x}) = \alpha\cdot\vec{0} = \vec{0} \Rightarrow A(\alpha\vec{x}) = 0 . \alpha\vec{x} \in N(A)$

Theorem Let A be an $m \times n$ matrix. Then $N(A)$ is a subspace of $\mathbf{R}^n$.

Theorem Let A be an $m \times n$ matrix. Then $N(A)$ is a subspace of $\mathbf{R}^n$.

Proof

(i) The set $N(A)$ is nonempty, since $\mathbf{0} \in N(A)$.

(ii) Let $\mathbf{x} \in N(A)$ and $\alpha \in \mathbf{R}$. Then $A\mathbf{x} = \mathbf{0}$ and

$$A(\alpha\mathbf{x}) = \alpha A\mathbf{x} = \alpha\mathbf{0} = \mathbf{0}.$$

So, $\alpha\mathbf{x} \in N(A)$.

(iii) Let $\mathbf{x}$ and $\mathbf{y}$ are elements of $N(A)$. Then $A\mathbf{x} = \mathbf{0}$ and $A\mathbf{y} = \mathbf{0}$. So

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

and $\mathbf{x} + \mathbf{y} \in N(A)$.

It then follows that $N(A)$ is a subspace of $\mathbf{R}^n$.

Example Let $A = \begin{pmatrix} 1 & -2 & 1 & 1 & 2 \\ -1 & 3 & 0 & 2 & -2 \\ 0 & 1 & 1 & 3 & 4 \\ 1 & 2 & 5 & 13 & 5 \end{pmatrix}$. Find the null space $N(A)$ of A.

Solution The reduced row echelon form of A is $\begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. So,

$Ax = 0$

$\rightarrow \begin{bmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = 0$

Example Let $A = \begin{pmatrix} 1 & -2 & 1 & 1 & 2 \\ -1 & 3 & 0 & 2 & -2 \\ 0 & 1 & 1 & 3 & 4 \\ 1 & 2 & 5 & 13 & 5 \end{pmatrix}$. Find the null space $N(A)$ of A.

Solution The reduced row echelon form of A is $\begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. So,

$$N(A) = \{\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)^T : A\mathbf{x} = \mathbf{0}\}$$

$$= \left\{ (x_1, x_2, x_3, x_4, x_5)^T : \begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$= \left\{ \begin{pmatrix} -3\alpha - 7\beta \\ -\alpha - 3\beta \\ \alpha \\ \beta \\ 0 \end{pmatrix} \right\} = \left\{ \alpha \begin{pmatrix} -3 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -7 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Definition (Linear Combination) Let $\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}$ be vectors in a vector space V.

$$y = \alpha_1\mathbf{v_1} + \alpha_2\mathbf{v_2} + \cdots + \alpha_n\mathbf{v_n}, \qquad \alpha_i \in \mathbf{R}$$

is called a **linear combination** of $\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}$.

Definition (Span) The set of all linear combinations of $\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}$

$$\{\alpha_1\mathbf{v_1} + \alpha_2\mathbf{v_2} + \cdots + \alpha_n\mathbf{v_n} : \quad \alpha_1, \alpha_2, \cdots, \alpha_n \in \mathbf{R}\}$$

is called the **span** of $\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}$ and denoted by

$$\text{Span}\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}.$$

Theorem If $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ are elements of a vector space V, then $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a subspace of V.

Proof

(i) $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ contains $\mathbf{v}_1$ and hence nonempty.

(ii) Let β be a scalar and let $\mathbf{v} = \alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n$ be an arbitrary element of $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$. Since

$$\beta\mathbf{v} = (\beta\alpha_1)\mathbf{v}_1 + \cdots + (\beta\alpha_n)\mathbf{v}_n,$$

it follows that $\beta\mathbf{v} \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$.

(iii) Let $\mathbf{v} = \alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n$ and $\mathbf{w} = \beta_1\mathbf{v}_1 + \cdots + \beta_n\mathbf{v}_n$. Then

$$\mathbf{v} + \mathbf{w} = (\alpha_1 + \beta_1)\mathbf{v}_1 + \cdots + (\alpha_n + \beta_n)\mathbf{v}_n.$$

So, $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a subspace of V.

Definition (Spanning set) The set $\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$ is a **spanning set** for V if and only if every vector in V can be written as a linear combination of $\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$.

1 vector. 2 and more. be arbitrary $\mathbf{y} \in V$. vector space.

To prove $\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$ spans a vector space V:

Step 1 For **arbitrary** $\mathbf{y} \in V$, consider the system

$$\mathbf{y} = a_1\mathbf{v_1} + a_2\mathbf{v_2} + \cdots + a_n\mathbf{v_n}, \quad a_i \in \mathbf{R}$$

Linear combination

Step 2 **Solve** for $a_1, a_2, \cdots, a_n$

- If **consistent**, then $\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$ spans V.
- If **inconsistent**, then $\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$ can not span V!

→ find out least one sol

Example Is $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}\right\}$ a spanning set for $\mathbf{R}^2$?

Solution Let $\begin{pmatrix}a\\b\end{pmatrix} \in \mathbf{R}^2$. Solve $\alpha_1 \begin{pmatrix}1\\0\end{pmatrix} + \alpha_2 \begin{pmatrix}2\\0\end{pmatrix} = \begin{pmatrix}a\\b\end{pmatrix}$. Since there is no solution of $\begin{cases}\alpha_1 + 2\alpha_2 = a\\ 0 = b\end{cases}$, $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}\right\}$ is not a spanning set for $\mathbf{R}^2$.

Example Is $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$ a spanning set for $\mathbf{R}^2$?

Solution Let $\begin{pmatrix}a\\b\end{pmatrix} \in \mathbf{R}^2$. Solve $\alpha_1 \begin{pmatrix}1\\0\end{pmatrix} + \alpha_2 \begin{pmatrix}2\\0\end{pmatrix} + \alpha_3 \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}a\\b\end{pmatrix}$, we have $\begin{cases}\alpha_1 = a - 2\beta\\ \alpha_2 = \beta\\ \alpha_3 = b\end{cases}$, where β is any real number. So $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$ is a spanning set for $\mathbf{R}^2$.

Example Let P_3 be the set of all polynomials of degree less than 3. Is $\{x^2, x + x^2, -x - x^2\}$ a spanning set of P_3?

$P_n = a_0 + a_1 x^1 + a_2 x^2 + \dots + a_{n-1} x^{n-1}$

$\rightarrow P_3 = a_0 + a_1 x^1 + a_2 x^2$

Example Let P_3 be the set of all polynomials of degree less than 3. Is $\{x^2, x + x^2, -x - x^2\}$ a spanning set of P_3?

3 vectors

Solution

Let $ax^2 + bx + c \in P_3$. We wish to find $\alpha_1, \alpha_2, \alpha_3$ such that

$\alpha_1 v_1$ $\alpha_2 v_2$ $\alpha_3 v_3$

$$\begin{aligned} ax^2 + bx + c &= \alpha_1(x^2) + \alpha_2(x + x^2) + \alpha_3(-x - x^2) \\ &= (\alpha_1 + \alpha_2 - \alpha_3)x^2 + (\alpha_2 - \alpha_3)x \end{aligned}$$

Setting $\begin{cases} \alpha_1 + \alpha_2 - \alpha_3 = a \\ \alpha_2 - \alpha_3 = b \\ 0 = c \end{cases}$.

The system is consistent only if $c = 0$. The system is inconsistent if $c \neq 0$. So $\{x^2, x + x^2, -x - x^2\}$ is not a spanning set for P_3.

Matrix $\left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 1 & 1 & 0 & b \\ 1 & 0 & 0 & c \end{array}\right]$

Extra Exercises*

Which of the following are spanning sets for $\mathbf{R}^3$?

(a) $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ in $\mathbf{R}^3$.

(b) $\{\mathbf{i}, \mathbf{j}, \mathbf{k}, (1,2,3)^T\}$ in $\mathbf{R}^3$.

(c) $\{(1,1,1)^T, (1,1,0)^T, (1,0,0)^T\}$. →

(d) $\{(1,0,1)^T, (0,1,0)^T\}$



$(a,b,c)^T = \alpha_1(1,1,1)^T + \alpha_2(1,1,0)^T + \alpha_3(1,0,0)^T$

$a = \alpha_1 + \alpha_2 + \alpha_3$

$b = \alpha_1 + \alpha_2 \Rightarrow$

$c = \alpha_1$

$\begin{cases} \alpha_1 = c \\ \alpha_2 = b - c \\ \alpha_3 = a - b \end{cases}$ yes

⇒ Square matrix → { nonsingular → det(A) ≠ 0 → consistent
singular → det(A) = 0 → inconsistent

Chapter 3 Vector Spaces

Section 3.3 Linear independence

辅助定理.

Lemma Given n vectors $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$, it is possible to write one of the vectors as a linear combination of the other $n-1$ vectors if and only if there exist scalars c_i, $i = 1, \cdots, n$, not all zero, such that

只要有一个 c_i 不等于0

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}.$$

就可以将这个表达出来

Proof Suppose

$$\mathbf{v}_n = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_{n-1}\mathbf{v}_{n-1}. \quad \neq 0$$

Subtracting $\mathbf{v}_n$ from both sides of this equation, we get

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_{n-1}\mathbf{v}_{n-1} - \mathbf{v}_n = \mathbf{0}.$$

Conversely, if $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$ and at least one of the c_i's, say c_n, is nonzero, then

$$\mathbf{v}_n = \big(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_{n-1}\mathbf{v}_{n-1}\big)/c_n$$

Definition (Linearly independent) The vectors $\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}$ in a vector space V are said to be **linearly independent** if

$$c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_n\mathbf{v_n} = \mathbf{0},$$

implies that all the scalars $c_i = 0$, $i = 1, 2, \cdots, n$.

Definition (Linearly dependent) The vectors $\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}$ in a vector space V are said to be **linearly dependent** if there exist scalars c_i ,$i = 1, 2, \cdots, n$, not all zero, such that $c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_n\mathbf{v_n} = \mathbf{0}$.

To check whether $\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$ are linearly independent:

Step 1 Consider the system

$$c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_n\mathbf{v_n} = \mathbf{0}.$$

Step 2 Solve for $c_1, c_2, \cdots, c_n$.

Homogenious

- If there is only the trivial solution (i.e. $c_i = 0$ for all i), then the vectors are linearly independent.
- If there are Infinitely many solutions, then the vectors are linearly dependent.

Example Are $(1,2,4)^T, (2,1,3)^T, (4,-1,1)^T$ linearly independent in $\mathbf{R}^3$?

Solution

If $c_1\begin{pmatrix}1\\2\\4\end{pmatrix} + c_2\begin{pmatrix}2\\1\\3\end{pmatrix} + c_3\begin{pmatrix}4\\-1\\1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$, then $\begin{cases} 1c_1 & +2c_2 & +4c_3 & = 0 \\ 2c_1 & +1c_2 & -1c_3 & = 0 \\ 4c_1 & +3c_2 & +1c_3 & = 0 \end{cases}$.

$$\left(\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 2 & 1 & -1 & 0 \\ 4 & 3 & 1 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & -3 & -9 & 0 \\ 0 & -5 & -15 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & -3 & -9 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The solutions are $\begin{pmatrix}2\alpha\\-3\alpha\\\alpha\end{pmatrix}$ where α is any real number.

(If $\alpha = 1$, then $\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix} = \begin{pmatrix}2\\-3\\1\end{pmatrix}$ and so $2\begin{pmatrix}1\\2\\4\end{pmatrix} - 3\begin{pmatrix}2\\1\\3\end{pmatrix} + \begin{pmatrix}4\\-1\\1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$.)

Therefore, the three vectors are linearly dependent.

Example Is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ linearly independent?

Solution Solving $c \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, we have $c = 0$. So $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is linearly independent.

Example Are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ linearly independent?

Solution Solving $c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, we have $c_1 = c_2 = 0$. So $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly independent.

Example Are $\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}$ linearly independent?

Solution Solving $c_1\begin{pmatrix}1\\0\end{pmatrix} + c_2\begin{pmatrix}2\\0\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$, we have $c_1 = -2\alpha$, $c_2 = \alpha$, where α is a real number. So $\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}$ are linearly dependent.

Example Are $\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}$ linearly independent?

Solution Solving $c_1\begin{pmatrix}1\\0\end{pmatrix} + c_2\begin{pmatrix}2\\0\end{pmatrix} + c_3\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$, we have $c_1 = -2\alpha$, $c_2 = \alpha$ and $c_3 = 0$, where α is a real number. So $\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}$ are linearly dependent.

Example Show that $(x-1)(x-2), (x-1), 1$ are linearly independent in P_3.

Solution Suppose

$$a_1(x-1)(x-2) + a_2(x-1) + a_3 = 0.$$

Then

$$a_1x^2 + (-3a_1 + a_2)x + (2a_1 + a_3) = 0.$$

Comparing the coefficients on both sides, we have

$$\begin{cases} a_1 = 0 \\ -3a_1 + a_2 = 0 \\ 2a_1 + a_3 = 0 \end{cases}, \quad \text{which gives} \quad \begin{cases} a_1 = 0 \\ a_2 = 0 \\ a_3 = 0 \end{cases}.$$

Hence, $(x-1)(x-2), (x-1), 1$ are linearly independent in P_3.

Square Matrix

Theorem Let $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$ be n vectors in $\mathbf{R}^n$ and let $X = (\mathbf{x}_1, \cdots, \mathbf{x}_n)$. The vectors $\mathbf{x}_1, \cdots, \mathbf{x}_n$ will be linearly dependent if and only if X is singular.

Proof The equation

$X \cdot C = 0$

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_n\mathbf{x}_n = \mathbf{0} \qquad c \neq 0$$

can be rewritten as a matrix equation $X\mathbf{c} = \mathbf{0}$. This equation will have a nontrivial solution if and only if X is singular. Thus, $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$ will be linearly dependent if and only if X is singular.

if we want $c \neq 0$

$\Rightarrow \det(X) = 0$

X is singular

Theorem Let $\mathbf{v}_1, \cdots, \mathbf{v}_n$ be vectors in a vector space V.
The following are equivalent.

1. Any vector $\mathbf{v} \in \text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ can be written uniquely as a linear combination of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.
2. $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent.

(Negation version) The following are equivalent.

Not 1. There is a vector $\mathbf{v} \in \text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ can be written as more than one linear combinations of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.

Not 2. $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent.

A $\Leftrightarrow$ B and not B $\Leftrightarrow$ not A are equivalent

(Negation version) The following are equivalent.

Not 1. There is a vector $\mathbf{v} \in \text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ can be written as more than one linear combinations of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.

Not 2. $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent.

Proof of "Not (1) $\Rightarrow$ Not (2)"

If $\mathbf{x} \in \text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$. Suppose that $\mathbf{x}$ have two expressions of linear combination

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n \qquad (1)$$

$$\mathbf{x} = d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \cdots + d_n\mathbf{v}_n \qquad (2)$$

If $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ are linearly independent, then subtracting (2) from (1) yields

$$(c_1 - d_1)\mathbf{v}_1 + (c_2 - d_2)\mathbf{v}_2 + \cdots + (c_n - d_n)\mathbf{v}_n = \mathbf{0}. \qquad (3)$$

By the linear independence of $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ the coefficients of (3) must all be 0. Hence, $c_i = d_i$, $i = 1, \cdots, n$.

(Negation version) The following are equivalent.

Not 1. There is a vector $\mathbf{v} \in \text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ can be written more than one linear combinations of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.

Not 2. $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent.

Proof Not "(2) $\Rightarrow$ Not (1)" If $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ are linearly dependent, then there exist $d_1, \cdots, d_n$, not all 0, such that

$$d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + \cdots + d_n\mathbf{v}_n = \mathbf{0}. \qquad (4)$$

Adding (1) and (4), we get

$$\mathbf{x} = (c_1 + d_1)\mathbf{v}_1 + \cdots + (c_n + d_n)\mathbf{v}_n.$$

Since the d_i's are not all 0, thus, if $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent, the representation of a vector as a linear combination of $\mathbf{v}_1, \cdots, \mathbf{v}_n$ is not unique.

Theorem Let $\mathbf{v}_1, \cdots, \mathbf{v}_n$ be vectors in a vector space V.
The following are equivalent.

1. Any vector $\mathbf{v} \in \text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ can be written uniquely as a linear combination of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.
2. $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent.

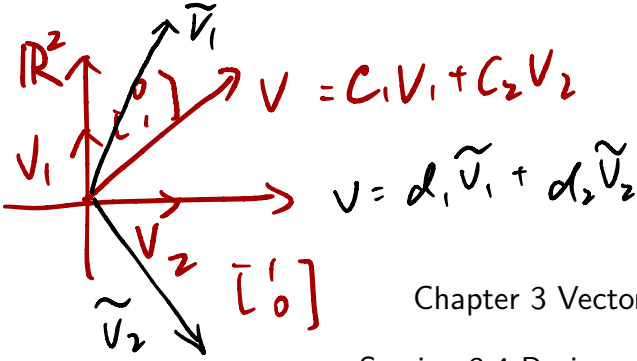
So, if $\text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\} = V$, The following are equivalent.

1. A vector $\mathbf{v} \in V$ can be written uniquely as a linear combination of $\mathbf{v}_1, \cdots, \mathbf{v}_n$.
2. $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent.

If a) $\text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_n\} = V$ and b) $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent, then for any vector $\mathbf{v} \in V$,

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$$

for some unique c_i. And we will call $(c_1, c_2, \cdots, c_n)$ the coordinates of $\mathbf{v}$.



Chapter 3 Vector Spaces

Section 3.4 Basis and Dimension

Example Re-visit

There are infinitely many sets of vectors that span the same vector space! e.g.

$$\mathbf{R}^3 = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, (1,2,3)^T\} = \cdots$$

Basis : 3 vectors

What is the minimal spanning set for a vector space V?

By "minimal", not too many vectors in a spanning set, also not too few.

Answer: Basis!

Definition (Basis, plural form: bases) The vectors $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ form a *basis* of a vector space V if

1. $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ are linearly independent, and
2. $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a spanning set for V.

In this case, $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a *basis* of V.

Example $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis of $\mathbf{R}^3$.

Reason

1. If $c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, then $c_1 = c_2 = c_3 = 0$.
2. For any vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbf{R}^3$, if $\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, then $\begin{cases} \alpha_1 = a \\ \alpha_2 = b \\ \alpha_3 = c \end{cases}$.

Example Is $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, (1,1,1)^T\}$ a basis of $\mathbf{R}^3$?
Solution Solving $c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + c_3\mathbf{e}_3 + c_4(1,1,1)^T = \mathbf{0}$, we get
$(c_1, c_2, c_3, c_4) = (-\alpha, -\alpha, -\alpha, \alpha)$.
Hence, $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, (1,1,1)^T$ are linearly dependent. (有非0解)
So, $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, (1,1,1)^T\}$ is not a basis of $\mathbf{R}^3$.

Example Is $\{\mathbf{e}_1, \mathbf{e}_2\}$ a basis of $\mathbf{R}^3$?
Solution
Let $(a,b,c)^T \in \mathbf{R}^3$. Solving $\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 = (a,b,c)^T$, we have $\begin{cases} \alpha_1 = a \\ \alpha_2 = b \\ 0 = c \end{cases}$.

Since $0 = c$ is inconsistent when $c \neq 0$, there is no solution for $\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 = (a,b,c)^T$ when $c \neq 0$.
Since $\{\mathbf{e}_1, \mathbf{e}_2\}$ is not a spanning set of $\mathbf{R}^3$, $\{\mathbf{e}_1, \mathbf{e}_2\}$ is not a basis of $\mathbf{R}^3$.

Example Is $\{\mathbf{e}_1, \mathbf{e}_2, (1,1,1)^T\}$ a basis of $\mathbf{R}^3$?

Solution

1 If $c_1\begin{pmatrix}1\\0\\0\end{pmatrix}+c_2\begin{pmatrix}0\\1\\0\end{pmatrix}+c_3\begin{pmatrix}1\\1\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$, then $\begin{pmatrix}c_1+c_3\\c_2+c_3\\c_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$ and so $c_1 = c_2 = c_3 = 0$. So, $\mathbf{e}_1, \mathbf{e}_2, (1,1,1)^T$ are linearly independent.

2. Let $(a,b,c)^T \in \mathbf{R}^3$. Solving $\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \alpha_3(1,1,1)^T = (a,b,c)^T$, we have $\begin{cases}\alpha_1 = a - c\\ \alpha_2 = b - c\\ \alpha_3 = c\end{cases}$. $\{\mathbf{e}_1, \mathbf{e}_2, (1,1,1)^T\}$ is a spanning set for $\mathbf{R}^3$.

Conclusion Since $\mathbf{e}_1, \mathbf{e}_2, (1,1,1)^T$ are linearly independent and form a spanning set for $\mathbf{R}^3$, they form a basis of $\mathbf{R}^3$.

$P_n = \{c_0 + c_1 x + \cdots + c_{n-1} x^{n-1} \mid c_i \in \mathbb{R}\}$

① $c_1 x + c_2 x^2 + \cdots + c_{n-1} x^{n-1} = 0$ } Linear Independence

show $\Rightarrow c_i = 0$

Example (Standard basis) $\{\mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_n\}$ is called the standard basis of $\mathbf{R}^n$.

Example (Standard basis) $\{1, x, x^2, \cdots, x^{n-1}\}$ is called the standard basis of P_n.

Example (Standard basis) $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ is the standard basis of the vector space $\mathbf{R}^{2\times 2}$ of 2×2 matrices.

proof : ① $C_1 + C_2 x^1 + \dots + C_{n-1} x^{n-1} = 0$

Let $x > 0 \Rightarrow C_1 = 0$

$C_2 x + \dots + C_{n-1} x^{n-1} = 0$

$x(C_2 + \dots + C_{n-1} x^{n-2}) = 0$

For $A \in \mathbb{R}^{m \times n}$ (matrix) $\quad null(A) = \{x \in \mathbb{R}^n : Ax = 0\}$: subspace of $\mathbb{R}^n$

Thm Let $V = span\{v_1, \dots, v_n\}$ ⟩ 如果 n 个向量为 V 的张成集

Let $u_i \in V$, $i = 1, \dots m$. $m > n$ $\Rightarrow \dim V \le n$

If $m > n$. then $\{u_1, \dots, u_m\}$ is linear dependent.

if $m > n$. and $\{u_1, \dots, u_m\} \in V$

$\{u_1, \dots, u_m\}$ must be linear dependent. 线性相关

pf : Since $u_i \in V$. $\exists\ a_{ij}$, $j = 1, \dots, n$ (系数)

s.t. $u_i = \sum_{j=1}^{n} a_{ij} v_j$, Let $c_i \in \mathbb{R}$

$$\sum_{i=1}^{n} c_i u_i = \sum_{i=1}^{n} c_i \left(\sum_{j=1}^{n} a_{ij} v_j\right)$$

$$= \sum_{i,j} c_i a_{ij} v_j = \sum_{j=1}^{n} \sum_{i=1}^{m} c_i a_{ij} v_j$$

$$= \sum_{j=1}^{n} v_j \underbrace{\left(\sum_{i=1}^{m} a_{ij} c_i\right)}_{d_j}$$

$$d_j = \sum_{i=1}^{m} a_{ij} c_i \quad , \quad j = 1, \dots, n$$

$$\begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & \vdots & & \vdots \\ \vdots & & & \\ a_{1m} & \cdots & & a_{mn} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix}$$

if $d_i = 0$. $\forall i$

$m > n \Rightarrow \exists$ nontrivial solution 为 $\underline{c}$

$\Rightarrow \sum_{i=1}^{m} c_i u_i = 0$. c_i all not all zero $\Rightarrow \{u_i\}_{i=1}^{m}$ is nearly dependent.

Corollary: If $\{v_i\}_{i=1}^n$ and $\{u_j\}_{j=1}^m$ are basis for V.

then $m = n$

$\dim V = n$ (unique)

Pf: If $m > n$ (Contradiction) · by previous theorem

$\{u_i\}_{i=1}^m$ is linearly dependent, but $\{u_i\}_{i=1}^m$ is a basis for V.

Contradiction!

Similarly, $n > m$ can't be true. So $m = n$

Thm: Suppose $\dim V = n$

Then $\{v_i\}_{i=1}^n$ is linearly indept iff $\text{span}\{v_i\}_{i=1}^n = V$

pf: Since $\dim V = n$. $\{w_i\}_{i=1}^n$ is a basis for V

$\Rightarrow \text{span}\{w_i\}_{i=1}^n = V$

$\Rightarrow$ the set $\{v_1 \dots v_n, v\}$ is linearly dept.

more than n vectors

$\Rightarrow \exists\ c_i$ not all zero s.t. $c_1 v_1 + \dots + c_n v_n + c_{n+1} v = 0$

In fact $c_{n+1} \neq 0$

(If $c_{n+1} = 0$, then $c_i \neq 0$ for some $i = 1, \dots, n$

$\Rightarrow c_1 v_1 + \dots + c_n v_n = 0 \Rightarrow \{v_i\}_{i=1}^n$ is linearly dept.

Contradiction!)

$c_{n+1} \neq 0$

$\Rightarrow v = -\frac{1}{c_{n+1}} \sum_{i=1}^n c_i v_i \Rightarrow v \in \text{span}\{v_i\}_{i=1}^n$

Let $v \in V$.

Since $v = \sum_{i=1}^n \left(-\frac{c_i}{c_{n+1}}\right) v_i$

Linear combination

= ⇐ " Suppose $\{v_i\}_{i=1}^n$ is linearly dep. (Contradiction)

Then $\exists\ c_i$, not all zero, s.t. $\sum_{i=1}^{n} c_i v_i = 0$

WLOG, Let $c_n \neq 0$

(不失一般性)

$$\Rightarrow v_n = -\frac{1}{c_n}\sum_{i=1}^{n} c_i v_i = \sum_{i=1}^{n}\left(-\frac{c_i}{c_n}\right)v_i$$

$\Rightarrow$ Span$\{v_1, \dots v_{n-1}\} = V$ (Since $v = \sum_{i=1}^{n}\left(-\frac{c_i}{c_n}\right)v_i$, can be written in linear combination)

If $\{v_i\}_{i=1}^{n-1}$ is linearly indep.

Then it's a basis for V

$\Rightarrow$ dim$V = n-1$, Contradiction!

Else, if $\{v_i\}_{i=1}^{n-1}$ is linearly dep.

then repeat the previous process until we get a set $\{v_1, \dots v_k\}$ is linearly indep. ($k < n$)

and dim$V = k$. Contraction!

Thm 3 Suppose dim$V = n$.

① not set of fewer than n vectors spans V.

② less than n linearly indepent vectors can be extended to form a basis of V.

③ Any set of more than n vectors that span V can be pared down (cut) to a basis of V

(Contradiction)

Pf of (1): Let $v_i \in V$. $i = 1, \dots k$. $k < n$

Suppose Span$= \{v_1, \dots v_k\} = V$

(a) linearly indep. then it's a basis, dim$V = k < n$. Contradiction!

(b) linearly dept. then we can find a subset of it which spans V but linearly Indept.

Then $\dim V = k < n$ Contradiction!

pf: (2) Let $\{v_1, \dots, v_k\}$ $k<n$. be linearly Indept.

By (1) $\text{Span}\{v_1, \dots v_k\} \neq V$

, is a proper subset of V (不能代表 V. 有些元素都在 V 中)

真子集

So $\exists\, v \in V$, s.t. $v \notin \text{span}\{v_1, \dots, v_k\}$

use

Claim: $\{v_1, \dots, v_k, v\}$ is linearly indept. $\Rightarrow$ Contradiction to prove

pf: If it's linearly dept. then $\exists c_1, \dots, c_{k+1}$ not all zero. s.t.

$$c_1 v_1 + \dots + c_k v_k + c_{k+1} v = 0$$

but $c_{k+1} = 0$

$\Rightarrow c_1 = \dots = c_{k+1} = 0$ Contradiction!

(Since linearly indept.)

pf: (3) Suppose $\text{span}\{v_1, \dots v_m\} = V$. $m > n$

$\{v_1, \dots v_m\}$ is linearly dept.

So $\exists\, c_1, \dots c_m$, not all zero s.t.

$$c_1 v_1 + \dots + c_m v_m = 0$$

WLOG. Say $c_m \neq 0$. then $c_m = -\frac{1}{c_m}(c_1 v_1 + \dots + c_{m-1} v_{m-1})$

$\text{span}\{v_1 \dots v_{m-1}\} = V = \text{span}\{v_1, \dots, v_m\}$

Continue the process until we have a set of n vectors that spans V.

Thm 3 Suppose $\dim V = n$

and $v_i \in V$, $i = 1, \cdots, m$.

(1) If $m > n$

(a) $\{v_1, \cdots, v_m\}$ is linearly dep^t.

(b) $\text{span}\{v_1, \cdots, v_m\} = V$, then we can reduce it to form a basis of V

(2) If $m < n$.

(a) $\{v_1, \cdots, v_m\}$ can not span V.

(b) If $\{v_1, \cdots, v_m\}$ are linearly indept, we can extend it to a basis of V

(3) If $m = n$.

(a) If $\{v_1, \cdots, v_n\}$ are linearly indept. then it's a basis.

(b) If $\text{span}\{v_1, \cdots, v_n\} = V$, then it's a basis.

Example Let $A = \begin{pmatrix} 1 & -2 & 1 & 1 & 2 \\ -1 & 3 & 0 & 2 & -2 \\ 0 & 1 & 1 & 3 & 4 \\ 1 & 2 & 5 & 13 & 5 \end{pmatrix}$. Find a basis of the null space $N(A)$ of A.

Solution The reduced row echelon form of A is $\begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

$$\begin{aligned} N(A) &= \{\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)^T : A\mathbf{x} = \mathbf{0}\} \\ &= \left\{ (x_1, x_2, x_3, x_4, x_5)^T : \begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\} \\ &= \left\{ \begin{pmatrix} -3\alpha - 7\beta \\ -\alpha - 3\beta \\ \alpha \\ \beta \\ 0 \end{pmatrix} \right\} = \left\{ \alpha \begin{pmatrix} -3 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -7 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\} \end{aligned}$$

$\left\{ \begin{pmatrix} -3 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -7 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$ is a basis of $N(A)$.

Theorem 1 If $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a spanning set for V, then any collection of m vectors in V, $m > n$, is linearly dependent.

Corollary If both $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m\}$ are bases for a vector space V, then $n = m$.

Proof of Corollary Let both $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m\}$ be bases for V. Since $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ span V and $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m$ are linearly independent, it follows from the above theorem that $m \leq n$. By the same reasoning, $\{\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m\}$ span V and $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ are linearly independent, so $n \leq m$. The assertion of $m = n$ is proved.

Theorem 1 If $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a spanning set for V, then any collection of m vectors in V, $m > n$, is linearly dependent. 线性相关

Proof Let $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m$ be m vectors in V, where $m \geq n$. Since $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ span V, we have $\mathbf{u}_i = a_{i1}\mathbf{v}_1 + \cdots + a_{in}\mathbf{v}_n$ and a linear combination

$$
\begin{aligned}
c_1\mathbf{u}_1 + \cdots + c_m\mathbf{u}_m &= c_1 \sum_{j=1}^{n} a_{1j}\mathbf{v}_j + \cdots + c_m \sum_{j=1}^{n} a_{kj}\mathbf{v}_j \\
&= \sum_{i=1}^{m} \left[c_i \sum_{j=1}^{n} a_{ij}\mathbf{v}_j \right] \\
&= \sum_{j=1}^{n} \left[\sum_{i=1}^{m} a_{ij}c_i \right] \mathbf{v}_j.
\end{aligned}
$$

?

find the value

Proof (continuity) Now consider the system of equations with unknown c_i's as follows

$$\sum_{i=1}^{m} a_{ij} c_i = 0, j = 1, 2, \cdots, n.$$

{ m equations
n unknown

This is a homogeneous system with more unknowns than equations. Therefore, by a theorem in Sc 1.2 (An $m \times n$ homogeneous system of linear equations has a nontrivial solution if $m < n$.) the system must have a nontrivial solution, denoted by $c_1^*, \cdots, c_m^*$. But then

$$c_1^* \mathbf{u}_1 + \cdots + c_m^* \mathbf{u}_m = \sum_{j=1}^{n} 0 \mathbf{v}_j.$$

Hence, $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_m$ are linearly dependent.

Definition (Dimension) Let V be a vector space. If V has a basis consisting of n vectors, we say that V has *dimension* n and write $\dim V = n$.

Example $\mathbf{R}^n$ is of dimension n.

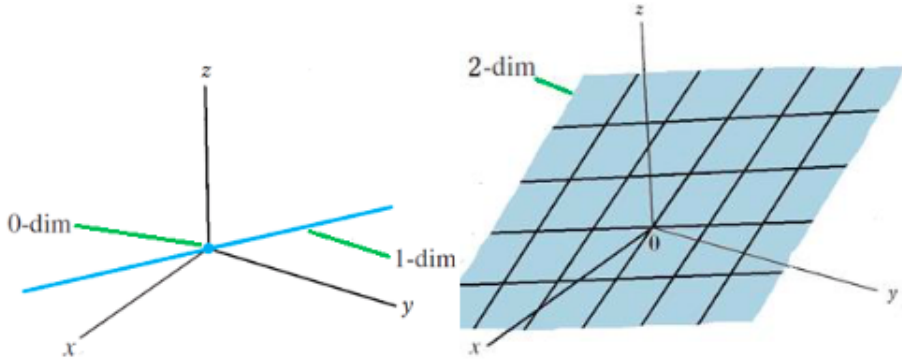
Example $\mathbf{R}^{m\times n}$ is of dimension mn.

Example P_n is of dimension n.

Example The subspace $\{\mathbf{0}\}$ of a vector space V is said to have dimension 0. (Some books would say that the basis of $\{\mathbf{0}\}$ is $\emptyset$.)

Example The subspaces of $\mathbf{R}^3$ can be classified by dimension.

- 0-dimensional subspaces: The zero subspace.
- 1-dimensional subspaces: Any subspace spanned by a single nonzero vector. Such subspaces are lines through the origin.
- 2-dimensional subspaces: Any subspace spanned by two linearly independent vectors. Such subspaces are planes through the origin.
- 3-dimensional subspaces: $\mathbf{R}^3$



affine space

↳ a space plus a number

Definition (Finite-dimensional / Infinite-dimensional) V is said to be *finite-dimensional* if a basis of V has only finitely many elements. Otherwise, V is said to be *infinite-dimensional*.

Example $\mathbf{R}^n$, $\mathbf{R}^{m\times n}$, P_n are all finite-dimensional.

Example $\{1, x, x^2, \cdots\}$ is a basis of vector space P of all polynomials. So P is of infinite-dimensional.

Example The vector space of all continuous functions is infinite-dimensional.

Example The vector space of infinite sequences is infinite-dimensional.

Theorem 2 If V is a vector space of dimension $n > 0$, then

(I) any set of n linearly independent vectors spans V;

(II) any n vectors that span V are linearly independent.

Proof of (I) Suppose that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent and $\mathbf{x}$ is any other vector in V. Since V has dimension n, it has a basis consisting of n vectors and these vectors span V. It follows that the set of $\{\mathbf{v}_1, \cdots, \mathbf{v}_n, \mathbf{x}\}$ must be linearly dependent. Thus, there exist scalars $c_1, \cdots, c_n, c_{n+1}$, not all zero, such that

$$c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n + c_{n+1}\mathbf{x} = \mathbf{0} \qquad (\sharp)$$

The scalar $c_{n+1} \neq 0$, otherwise from $(\sharp)$ $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent. Hence, $(\sharp)$ can be solved for $\mathbf{x}$:

$$\mathbf{x} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n,$$

where $\alpha_i = -c_i/c_{n+1}$. Since $\mathbf{x}$ is arbitrary, the assertion (I) follows.

Proof of (II) Suppose that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ span V. If $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent, then one of the $\mathbf{v}_i$'s, say, $\mathbf{v}_n$, can be written as a linear combination of the others. It follows from Theorem 1 that $\mathbf{v}_1, \cdots, \mathbf{v}_{n-1}$ will still span V. If $\mathbf{v}_1, \cdots, \mathbf{v}_{n-1}$ are linearly dependent, we can eliminate another vector and still have a spanning set. We can continue eliminating vectors in this way until we arrive at a linearly independent spanning set with $k < n$ elements. But this contradicts $\dim V = n$. Therefore, $\mathbf{v}_1, \cdots, \mathbf{v}_n$ must be linearly independent.

Theorem 3 If V is a vector space of dimension $n > 0$, then

(i) no set of fewer than n vectors can span V;

(ii) any subset of fewer than n linearly independent vectors can be extended to form a basis for V;

(iii) any spanning set containing more than n vectors can be pared down to form a basis for V.

Proof of (i) and (ii) Statement (i) follows by the same reasoning that was used to prove part (I) of Theorem 2.

To prove (ii), suppose that $\mathbf{v}_1, \cdots, \mathbf{v}_k$ are linearly independent with $k < n$. It follows from (i) that $\text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_k\}$ is a proper subspace of V, and hence there exists a vector $\mathbf{v}_{k+1} \in V$ that is not in $\text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_k\}$. It then follows that $\mathbf{v}_1, \cdots, \mathbf{v}_k, \mathbf{v}_{k+1}$ must be linearly independent. If $k + 1 < n$, then, in the same manner, $\mathbf{v}_1, \cdots, \mathbf{v}_k, \mathbf{v}_{k+1}$ can be extended to a set of $k + 2$ linearly independent vectors. This extension process may be continued until a set $\{\mathbf{v}_1, \cdots, \mathbf{v}_k, \mathbf{v}_{k+1}, \cdots, \mathbf{v}_n\}$ of n linearly independent vectors is obtained.

Proof of (iii) Suppose that $\mathbf{v}_1, \cdots, \mathbf{v}_m$ span V and $m > n$. Then, by Theorem 1, $\mathbf{v}_1, \cdots, \mathbf{v}_m$ must be linearly dependent. It follows that one of the vectors, say, $\mathbf{v}_m$, can be written as a linear combination of the others. Hence, if $\mathbf{v}_m$ is eliminated from the set, the remaining $m-1$ vectors will still span V. If $m-1 > n$, we can continue to eliminate vectors in this manner until we arrive at a spanning set containing n elements.

Extra Exercise* Determine if the following set of vectors form a basis for $\mathbf{R}^3$.

1. $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ in $\mathbf{R}^3$.
2. $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, (1,2,3)^T\}$ in $\mathbf{R}^3$.
3. $\{(1,0,0)^T, (0,1,1)^T\}$ in $\mathbf{R}^3$.

Answer:
(a). Yes, (b). No, (c). No

Extra Example* Let
$S = \{(2,-3,5)^T, (8,-12,20)^T, (1,0,2)^T, (0,2,-1)^T, (7,2,0)^T\}$.
$\text{Span}(S) = \mathbf{R}^3$. Find a basis of $\mathbf{R}^3$ providing S.

Answer:

1. Since $(8,-12,20)^T = 4(2,-3,5)^T$, then remove it

$$\text{Span}\{(2,-3,5)^T, (1,0,2)^T, (0,2,-1)^T, (7,2,0)^T\} = \mathbf{R}^3.$$

2. Since $(7,2,0)^T = 2(2,-3,5)^T + 3(1,0,2)^T + 4(0,2,-1)^T$, then remove it

$$\text{Span}\{(2,-3,5)^T, (1,0,2)^T, (0,2,-1)^T\} = \mathbf{R}^3.$$

3. Assume $\alpha_1(2,-3,5)^T + \alpha_2(1,0,2)^T + \alpha_3(0,2,-1)^T = (0,0,0)^T$, the only solution is $\alpha_1 = \alpha_2 = \alpha_3 = 0$, linear independent.
Therefore, $\{(2,-3,5)^T, (1,0,2)^T, (0,2,-1)^T\}$ is a basis for $\mathbf{R}^3$.

Chapter 3 Vector Spaces

Section 3.5 Change of basis

Theorem in Sc3.3 Let $\{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$ be a basis for a vector space V. Then for each $\mathbf{v} \in V$, there exists a *unique* set of scalar $c_1, c_2, \cdots, c_n \in \mathbf{R}$ such that

$$\mathbf{v} = c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_n\mathbf{v_n}.$$

Definition (Coordinate vector) Suppose $\beta = \{\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}\}$ be an **ordered basis** for a vector space V and $\mathbf{v} \in V$. If

$$\mathbf{v} = c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_n\mathbf{v_n}$$

for some $c_1, c_2, \cdots, c_n \in \mathbf{R}$, then

$$\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \in \mathbf{R}^n$$

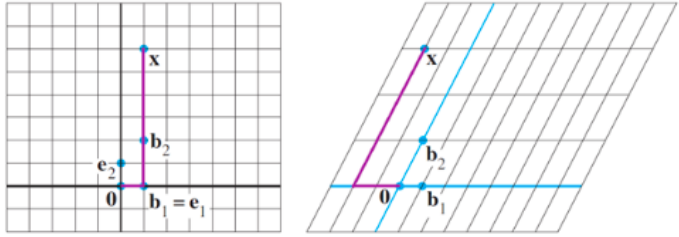
is called the **coordinate vector** of $\mathbf{v}$ with respect to β and is denoted $[\mathbf{v}]_\beta$. The c_i's are called the **coordinates** of $\mathbf{v}$ relative to β.

Example Consider two bases $\beta = \{\mathbf{u_1}, \mathbf{u_2}\}$ and $\gamma = \{\mathbf{e_1}, \mathbf{e_2}\}$ for $\mathbf{R}^2$, where $\mathbf{u_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{u_2} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Suppose $\mathbf{x} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$. Find $[\mathbf{x}]_\gamma$ and $[\mathbf{x}]_\beta$.

Solution Since $\mathbf{x} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 6\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \mathbf{e_1} + 6\mathbf{e_2}$, we have $[\mathbf{x}]_\gamma = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$.

Since $\mathbf{x} = -2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 1 \end{pmatrix} = -2\mathbf{u_1} + 3\mathbf{u_2}$, we have $[\mathbf{x}]_\beta = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

Different basis gives the same vector different coordinate vector.



Definition (Transition matrix) Let $\beta = \{\mathbf{u_1}, \cdots, \mathbf{u_n}\}$ and $\gamma = \{\mathbf{w_1}, \cdots, \mathbf{w_n}\}$ be two bases of a vector space V. The $n \times n$ matrix $\left([\mathbf{u_1}]_\gamma \,\middle|\, \cdots \,\middle|\, [\mathbf{u_n}]_\gamma \right)$ is called the **transition matrix** from β to γ.

In some books, the transition matrix from β to γ is denoted by $[I]_\beta^\gamma$ or $P_{\beta \to \gamma}$.

Example Consider two bases $\beta = \{\mathbf{u_1}, \mathbf{u_2}\}$ and $\gamma = \{\mathbf{e_1}, \mathbf{e_2}\}$ for $\mathbf{R}^2$, where $\mathbf{u_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{u_2} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Then the transition matrix from β to γ is $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

Theorem Let $\beta = \{\mathbf{u}_1, \cdots, \mathbf{u}_n\}$ and $\gamma = \{\mathbf{w}_1, \cdots, \mathbf{w}_n\}$ be two bases of a vector space V. Let $[I]_\beta^\gamma$ be the transition matrix from β to γ. Then

1. $[\mathbf{v}]_\gamma = [I]_\beta^\gamma [\mathbf{v}]_\beta$ for any $\mathbf{v} \in V$.
2. $[I]_\beta^\gamma$ is nonsingular and $\left([I]_\beta^\gamma\right)^{-1}$ is the transition matrix from γ to β.

Proof of (1) Let a_{ij} such that

$$\begin{aligned}
\mathbf{u}_1 &= a_{11}\mathbf{w}_1 + a_{21}\mathbf{w}_2 + \cdots + a_{n1}\mathbf{w}_n \\
\mathbf{u}_2 &= a_{12}\mathbf{w}_1 + a_{22}\mathbf{w}_2 + \cdots + a_{n2}\mathbf{w}_n \\
&\cdots \\
\mathbf{u}_n &= a_{1n}\mathbf{w}_1 + a_{2n}\mathbf{w}_2 + \cdots + a_{nn}\mathbf{w}_n
\end{aligned}$$

Let $\mathbf{v} \in V$. Suppose $[\mathbf{v}]_\beta = (c_1, c_2, \cdots, c_n)^T$. Then

$$\begin{aligned}
\mathbf{v} &= c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n \\
&= c_1(a_{11}\mathbf{w}_1 + a_{21}\mathbf{w}_2 + \cdots + a_{n1}\mathbf{w}_n) + \\
&\quad c_2(a_{12}\mathbf{w}_1 + a_{22}\mathbf{w}_2 + \cdots + a_{n2}\mathbf{w}_n) + \\
&\quad \cdots \\
&\quad c_n(a_{1n}\mathbf{w}_1 + a_{2n}\mathbf{w}_2 + \cdots + a_{nn}\mathbf{w}_n)
\end{aligned}$$

$$
\begin{aligned}
=&(c_1a_{11} + c_2a_{12} + \cdots + c_na_{1n})\mathbf{w}_1 + \\
&(c_1a_{21} + c_2a_{22} + \cdots + c_na_{2n})\mathbf{w}_2 + \\
&\vdots \\
&(c_1a_{n1} + c_2a_{n2} + \cdots + c_na_{nn})\mathbf{w}_n
\end{aligned}
$$

Thus

$$
[\mathbf{v}]_\gamma = \begin{pmatrix} c_1a_{11} + c_2a_{12} + \cdots + c_na_{1n} \\ c_1a_{21} + c_2a_{22} + \cdots + c_na_{2n} \\ \vdots \\ c_1a_{n1} + c_2a_{n2} + \cdots + c_na_{nn} \end{pmatrix}
$$

$$
= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = U[\mathbf{v}]_\beta
$$

Example Let $\beta = \{\mathbf{u}_1, \mathbf{u}_2\}$ and $\gamma = \{\mathbf{e}_1, \mathbf{e}_2\}$ be two ordered bases of $\mathbf{R}^2$ where $\mathbf{u}_1 = (2,1)^T, \mathbf{u}_2 = (-1,3)^T$. If the coordinate vector of $\mathbf{v}$ with respect to γ is $[\mathbf{v}]_\gamma = (5,-1)^T$, find its coordinate vector $[\mathbf{v}]_\beta$?

Solution The transition matrix from β to γ is $U = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$.

The transition matrix from γ to β is $U^{-1} = \frac{1}{7}\begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$.

So, $[\mathbf{v}]_\beta = U^{-1}[\mathbf{v}]_\gamma = \frac{1}{7}\begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

Let β and γ be two non-standard bases of a vector space V. If W is the transition matrix from to β to the standard basis and U^{-1} is the transition matrix from the standard basis to γ, then the transition matrix from β to γ is $U^{-1}W$.

Example Let $\beta = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \{(1,1,1)^T, (2,3,2)^T, (1,5,4)^T\}$ and $\gamma = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1,1,0)^T, (1,2,0)^T, (1,2,1)^T\}$.
(a) Which is the transition matrix from β to γ?
(b) Let $\mathbf{x} = 3\mathbf{u}_1 + 2\mathbf{u}_2 - 1\mathbf{u}_3$. Find the coordinates $[\mathbf{x}]_\gamma$ of $\mathbf{x}$ with respect to γ.

Solution (a) The transition matrix from β to γ is

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 5 \\ 1 & 2 & 4 \end{pmatrix} = \begin{pmatrix} -1 & 1 & -3 \\ -1 & -1 & 0 \\ 1 & 2 & 4 \end{pmatrix}$$

(b) The coordinates of $\mathbf{x}$ w.r.t. γ is

$$[\mathbf{x}]_\gamma = \begin{pmatrix} -1 & 1 & -3 \\ -1 & -1 & 0 \\ 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ -5 \\ 3 \end{pmatrix}.$$

F, G → bases. know $[v]_F$. what is $[v]_G$?

Sol: Find transition matrix

Let $F=\{u_1,\dots u_n\}$, $G=\{w_1,\dots w_n\}$ bases for $\mathbb{R}^n$

$\forall v \in \mathbb{R}^n$

$[v]_G = W^{-1}U[v]_F$

$[u_1\dots u_n]$

transition matrix

$$\begin{cases} U=[u_1|\cdots|u_n] \\ W=[w_1|\cdots|w_n] \end{cases}$$

Pf: $v=c_1u_1+\cdots+c_nu_n=d_1w_1+\cdots+d_nw_n \Rightarrow U\begin{bmatrix}c_1\\ \vdots\\ c_n\end{bmatrix}=W\begin{bmatrix}d_1\\ \vdots\\ d_n\end{bmatrix}$

$U\cdot[v]_F = W\cdot[v]_G \Rightarrow [v]_G = W^{-1}U[v]_F$.

$A\vec{x}=x_1a_1+\dots+x_na_n$

$A=[a_1|\cdots|a_n]$ (基向量)

Let

$E=\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}=u_1,\begin{bmatrix}2\\3\\2\end{bmatrix}=u_2,\begin{bmatrix}1\\5\\4\end{bmatrix}=u_3\right\}$

$F=\left\{\begin{bmatrix}1\\1\\0\end{bmatrix},\begin{bmatrix}1\\2\\0\end{bmatrix},\begin{bmatrix}1\\2\\1\end{bmatrix}\right\}$

be bases for $\mathbb{R}^3$.

$W^{-1}U[v]_E=[v]_F$

(1) What is the transition matrix from E to F?

(2) If $v=3u_1+2u_2-u_3$, what is $[v]_F$?

$\Rightarrow [v]_E=\begin{bmatrix}3\\2\\-1\end{bmatrix}$ $\begin{bmatrix}3\\2\\-1\end{bmatrix}$

Sol

(1) $W^{-1}U=\begin{bmatrix}1&1&1\\1&2&2\\0&0&1\end{bmatrix}^{-1}\begin{bmatrix}1&2&1\\1&3&5\\1&2&4\end{bmatrix}=\left[\begin{array}{ccc|ccc}1&1&1&1&2&1\\1&2&2&1&5&5\\0&0&1&1&2&4\end{array}\right]$

I

$\xrightarrow{R_2-R_1\to R_2}\left[\begin{array}{ccc|ccc}1&1&1&1&2&1\\0&1&1&0&1&4\\0&0&1&1&2&4\end{array}\right]$

$\xrightarrow[R_1-R_3\to R_1]{R_2-R_3\to R_2}\left[\begin{array}{ccc|ccc}1&0&0&1&1&-3\\0&1&0&-1&-1&0\\0&0&1&1&2&4\end{array}\right]\Rightarrow\begin{bmatrix}-1&-1&3\\-1&-1&0\\1&2&4\end{bmatrix}=W^{-1}U$

Transition Matrix

(1) Ans: $W^{-1}U[v]_E=[v]_F$